

Praktikum

Design of Wind Farms

Lecture 5:
NREL LandBOSSE / Financial Analysis

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Syllabus

No.	Week	Date (Wed)	9.30–10.15	10.30–11.15	11.30–12.15
1	14	15/04/26	Overview of Course Lecture 1: Background and Motivations	Lecture 2: Development of a Wind Farm Project	Lecture 3: Wake Behavior, Wind Farm Control Assignment 1: Basics and Recapitulation
2	15	22/04/26	Lecture 4: Wake Models, FLORIS	Assignment 1: Solution Assignment 2: Jensen's Wake Model and Wake Superposition in Wind Farms	Tutorial 1: FLORIS Installation and Setup
3	16	29/04/26	Lecture 5_1: LandBOSSE	Tutorial 2: LANDBOSSE Installation and Setup	Assignment 2: Solution Assignment 3: Wind Farm Analysis Using FLORIS
5	17	06/05/26	Lecture 5_2: Financial Analysis	Assignment 3: Solution	Assignment 4: Wind Farm Cost Analysis Using FLORIS + LandBOSSE)
6	18	13/05/26	Lecture 6: Layout Optimization	Assignment 4: Solution	Assignment 5: Layout Optimization
7	19	20/05/26	Assignment 5: Solution		Final group assignment: Introduction, Finalizing Groups, Q&A
8	20	27/05/26	Final group assignment: Design Process, Q&A, Writing Report		
9	21	03/06/26	No Class		
10	22	10/06/26	Final group assignment: Design Process, Q&A, Writing Report		
11	28	08/07/26	No Class Deadline: Electronic report submission until Tuesday, 07/07/2026, 23:59h		
12	29	15/07/26	Final presentation – 8:30h (Room 2701M)		



Agenda

- **NREL's Land-based Balance of System Systems Engineering (LandBOSSE)**
- **LandBOSSE Example**
- **Basics of financial analysis**



NREL's Land-based Balance of System Systems Engineering (LandBOSSE)

LandBOSSE model is a tool for modeling the balance-of-system (BOS) costs of land-based wind plants includes:

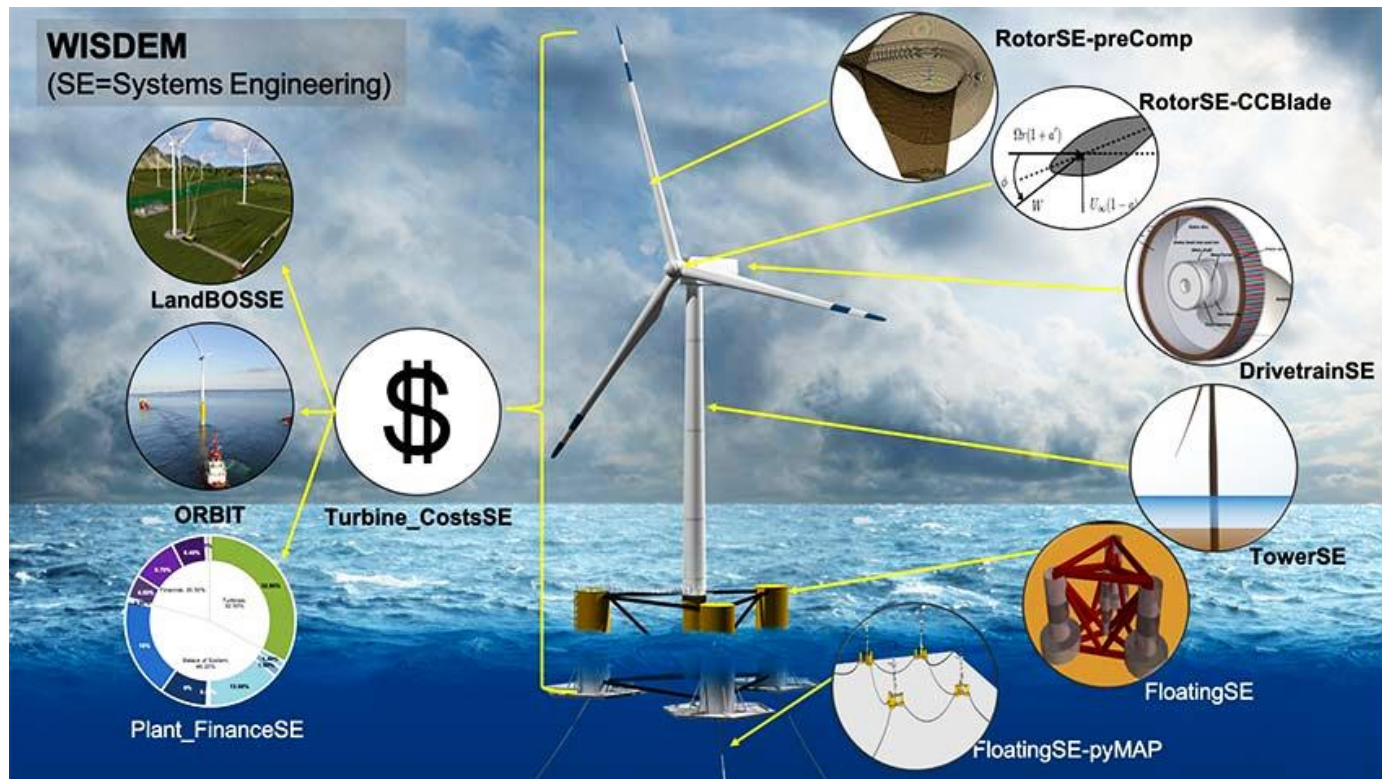
- All costs associated with installing a wind plant, such as permitting, labor, material, and equipment costs associated with site preparation, foundation construction, electrical infrastructure, and tower installation.
- LandBOSSE, however, was designed to help users explore tradeoffs between innovative design scenarios while balancing the level of detail and speed required for model execution.

Ref: [1]



Wind Plant Integrated Systems Design and Engineering Model (WISDEM)

However, LandBOSSE is a part of other software called WISDEM



LandBOSSE: Open-source on GitHub

Original code and documentation available at:

<https://github.com/WISDEM/LandBOSSE>

It can execute on macOS and Windows.

- Original version does not allow for irregular layouts and writes outputs to EXCEL
- ➔ **In this Praktikum we will work with an in-house further developed version of LandBOSSE!**
- ➔ **Available here:** <https://github.com/WindfarmDesigner/LandBOSSE>

Original documentation and more details available at:

<https://www.nrel.gov/docs/fy19osti/72201.pdf>



LandBOSSE: Installation and Operation Folder

- Download the source code and copy it in a folder of desire (or clone it).
- Open terminal, navigate to folder (using “cd”), type “pip install -e .”
Note: “-e” only installs a pointer and the source code remains in the original folder. If you prefer a copy to your environment, leave out “-e”.
- You might have to additionally install the *pytest* package (“pip install pytest”)
- Please use python version 3.10 – 3.12 (not tested above)
- In the terminal / your IDE, import and execute the function *run_landbosse* from *landbosse.main_function*.

Example:

```
import os
import numpy as np
from landbosse.main_function import run_landbosse

# Inputs
Turbine_coordinates = np.array([[0,0],[0,400],[0,800]])      # in [m]
Substation_coordinate = np.array([[200,400]])                # in [m]
Cable_Voltage = 30                                           # in [kV]. Available: 10,20,30,45,60 kV
WriteExcel = False                                           # set to 'True' if you additionally want EXCEL files as output
Display = True                                               # set to 'False' if you don't want landbosse prints

# Run
landbosse_results = run_landbosse(Turbine_coordinates, Substation_coordinate, Cable_Voltage, WriteExcel, Display)
```



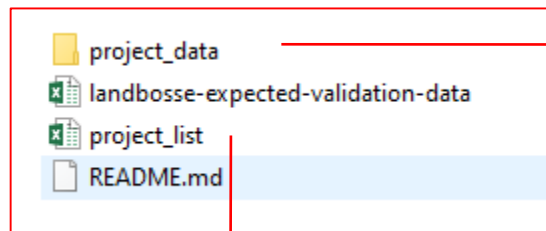
LandBOSSE Input and Output

- The **input** is comprised of three parts:

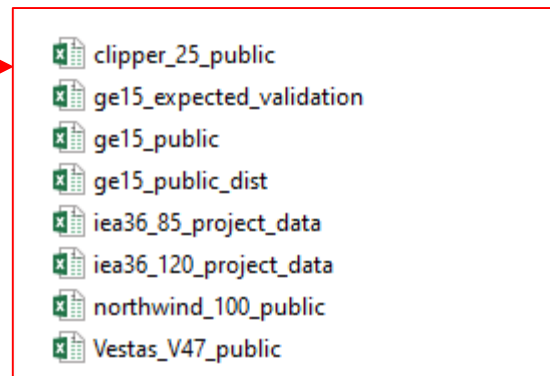
1. Array voltage level, and turbine and substation coordinates specified when calling `run_landbosse` in `landbosse.main_function`. Additionally, you need to specify if you want to write EXCEL output files and see prints.

```
landbosse_results = run_landbosse(Turbine_coordinates, Substation_coordinate, Cable_Voltage, WriteExcel, Display)
```

2. Project data, which are **.xlsx** spreadsheets that contain sheets of data about components, equipment, crews and so on. There is always at least one, and maybe more, files of project data.
3. A project list which combines parameters that define projects with reference to the project data those projects use. There is only one file that has the project list.



Main input parameters



Turbine cost data

Project ID	Project data file	Total project construction time	nTurbine rating MW	Hub height m	Rotor diameter m	Turbine spacing (times rotor diameter)	Row
foundation_validation_iea36_120	iea36_120_project_data	12	3.37	120	130		8

LandBOSSE Input Parameters

• Project list input

Project ID	Project data file	Total project construction time (n	Turbine rating MW	Hub height m	Rotor diameter m	Turbine spacing (times rotor diameter)	R
ge15_dist_01	ge15_public_dist	1	1.5	80	77		4
ge15_dist_05	ge15_public_dist	1	1.5	80	77		4
ge15_dist_10	ge15_public_dist	1	1.5	80	77		4
foundation_validation_ge15	ge15_public	9	1.5	80	77		4
foundation_validation_clipper25	clipper_25_public	9	2.5	80	96		4
foundation_validation_iea36_85	iea36_85_project_data	9	3.6	85	130		4
foundation_validation_iea36_120	iea36_120_project_data	9	3.6	120	130		4

Focus on:

- Total project construction time (months)
- Turbine rating MW
- Hub height m
- Rotor diameter m
- Fuel cost USD per gal
- Wind shear exponent
- Rated Thrust (N)
- Line Frequency (Hz)
- Distance to interconnect (miles)
- Interconnect Voltage (kV)

Ignore (no impact):

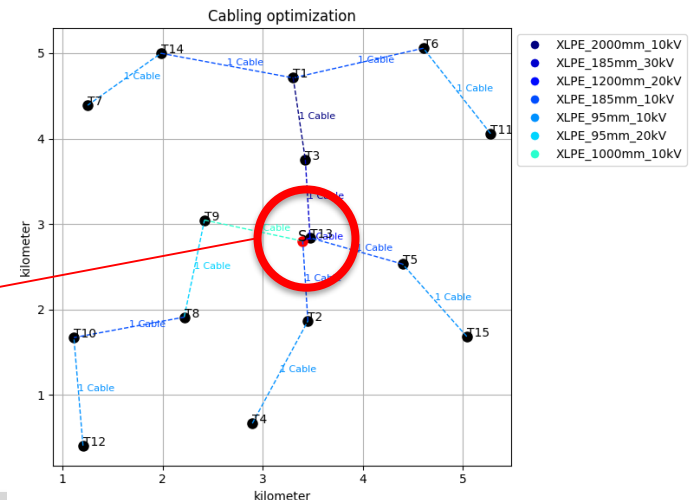
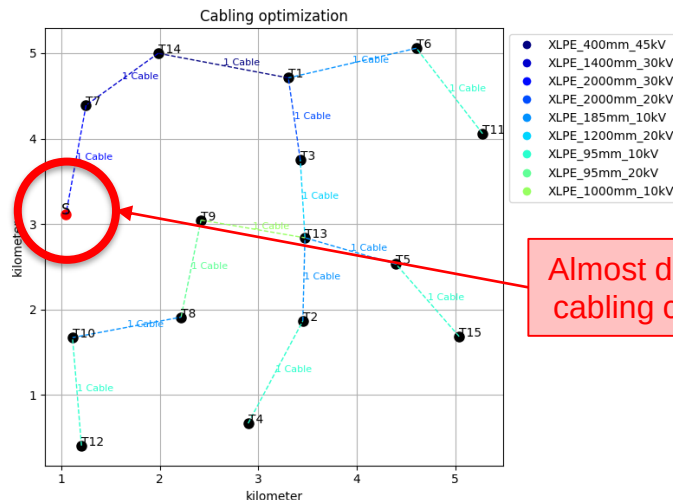
- Number of turbines
- Turbine spacing
- Row spacing
- Road length adder
- Flag for user-defined home run trench length
- Combined Homerun Trench Length to Substation (km)

Others: default values are good general guesses

LandBOSSE Input Parameters

Specific Layout Designs

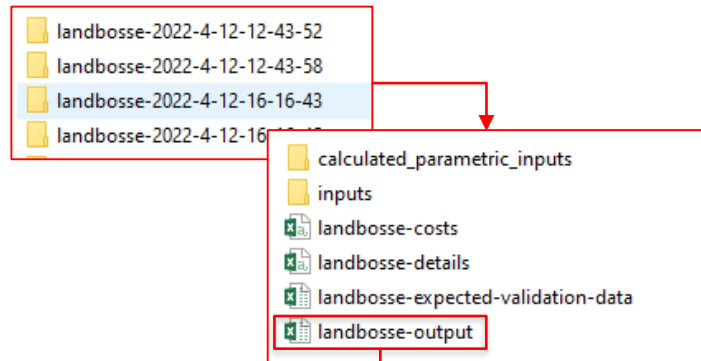
- Original LandBOSSE does not accommodate specific layout designs.
 - Internally developed code allowing for individual layouts → turbines and substation coordinates are model inputs
 - Before executing “run_landbosse”, update coordinates and specify array voltage
 - For now: Cabling optimization based on minimum-span-tree hence minimized total cable length (not cost)
- Care needs to be taken wrt substation! Must be considerably placed.



LandBOSSE Input and Output

- **Output** in python is a **dataframe** listing **all BoP costs** of the analysed project.
- Originally, landbosse only wrote EXCEL output files. You can still use this option by setting “WriteExcel = True”, although it is not recommended. Then, an output folder is created with a file structure comprised of two parts:
 - The actual data output from the model in the form of an .xlsx spreadsheet.
 - All the input files that created the output data.

Index	Project ID with serial	Number of turbines	Turbine rating MW	Rotor diameter m	Module	Type of cost	Cost per turbine	Cost per project	Cost per kW
12	ass4_1ea34	3	3.37	130	CollectionCost	Equipment rental	4167.74	12503.2	1.23672
13	ass4_1ea34	3	3.37	130	CollectionCost	Labor	19477.7	58433	5.77972
14	ass4_1ea34	3	3.37	130	CollectionCost	Materials	6885	20655	2.04303
15	ass4_1ea34	3	3.37	130	CollectionCost	Mobilization	2088.71	6266.14	0.619796
16	ass4_1ea34	3	3.37	130	DevelopmentCost	Equipment rental	0	0	0
17	ass4_1ea34	3	3.37	130	DevelopmentCost	Labor	50000	150000	14.8368
18	ass4_1ea34	3	3.37	130	DevelopmentCost	Materials	0	0	0
19	ass4_1ea34	3	3.37	130	DevelopmentCost	Mobilization	0	0	0
20	ass4_1ea34	3	3.37	130	DevelopmentCost	Other	0	0	0
21	ass4_1ea34	3	3.37	130	ErectionCost	Equipment rental	16008.2	48024.5	4.7502
22	ass4_1ea34	3	3.37	130	ErectionCost	Fuel	1922.03	5766.09	0.570335



Project ID with serial	Number of turbines	Turbine rating MW	Rotor diameter m	Module	Operation ID	Type of cost	Cost per turbine	Cost per project	USD/kW per project
Foundation_validation_1ea36	3	3.37	130	FoundationCost	Foundation	Equipment rental	\$ 10,799	\$ 32,128	\$ 3
Foundation_validation_1ea36	3	3.37	130	FoundationCost	Foundation	Labor	\$ 424,608	\$ 1,273,824	\$ 126
Foundation_validation_1ea36	3	3.37	130	FoundationCost	Foundation	Materials	\$ 199,084	\$ 597,253	\$ 59
Foundation_validation_1ea36	3	3.37	130	FoundationCost	Foundation	Mobilization	\$ 14,467	\$ 43,402	\$ 4
Foundation_validation_1ea36	3	3.37	130	SitePreparationCost	Roads	Materials	\$ 38,302	\$ 114,905	\$ 11
Foundation_validation_1ea36	3	3.37	130	SitePreparationCost	Roads	Equipment rental	\$ 21,317	\$ 63,950	\$ 6
Foundation_validation_1ea36	3	3.37	130	SitePreparationCost	Roads	Labor	\$ 101,034	\$ 303,103	\$ 30
Foundation_validation_1ea36	3	3.37	130	SitePreparationCost	Roads	Other	\$ 130,305	\$ 390,916	\$ 39
Foundation_validation_1ea36	3	3.37	130	SitePreparationCost	Roads	Mobilization	\$ 30,570	\$ 91,710	\$ 9
Foundation_validation_1ea36	3	3.37	130	SubstationCost	Substation	Other	\$ 0	\$ 0	\$ 0
Foundation_validation_1ea36	3	3.37	130	GridConnectionCost	Transmission and Distribution	Other	\$ 476,499	\$ 1,429,498	\$ 141
Foundation_validation_1ea36	3	3.37	130	CollectionCost	Collection	Equipment rental	\$ 4,515	\$ 13,545	\$ 1
Foundation_validation_1ea36	3	3.37	130	CollectionCost	Collection	Labor	\$ 21,083	\$ 63,250	\$ 6
Foundation_validation_1ea36	3	3.37	130	CollectionCost	Collection	Materials	\$ 12,795	\$ 38,386	\$ 4
Foundation_validation_1ea36	3	3.37	130	CollectionCost	Collection	Mobilization	\$ 2,427	\$ 7,280	\$ 1
Foundation_validation_1ea36	3	3.37	130	DevelopmentCost	Development	Equipment rental	\$ 0	\$ 0	\$ 0
Foundation_validation_1ea36	3	3.37	130	DevelopmentCost	Development	Labor	\$ 50,000	\$ 150,000	\$ 15
Foundation_validation_1ea36	3	3.37	130	DevelopmentCost	Development	Materials	\$ 0	\$ 0	\$ 0
Foundation_validation_1ea36	3	3.37	130	DevelopmentCost	Development	Mobilization	\$ 0	\$ 0	\$ 0
Foundation_validation_1ea36	3	3.37	130	DevelopmentCost	Development	Other	\$ 0	\$ 0	\$ 0
Foundation_validation_1ea36	3	3.37	130	ErectionCost	Erection	Equipment rental	\$ 14,061	\$ 42,183	\$ 4
Foundation_validation_1ea36	3	3.37	130	ErectionCost	Erection	Fuel	\$ 1,522	\$ 4,566	\$ 1
Foundation_validation_1ea36	3	3.37	130	ErectionCost	Erection	Labor	\$ 165,331	\$ 495,992	\$ 49
Foundation_validation_1ea36	3	3.37	130	ErectionCost	Erection	Mobilization	\$ 299,770	\$ 899,310	\$ 89
Foundation_validation_1ea36	3	3.37	130	ErectionCost	Erection	Other	\$ 0	\$ 0	\$ 0
Foundation_validation_1ea36	3	3.37	130	ErectionCost	Erection	Materials	\$ 0	\$ 0	\$ 0

LandBOSSE Output Parameters

- The **output** (dataframe and optional file in folder “output”)

- Foundation Cost
- Site Preparation Cost
- Sub station Cost
- Grid Connection Cost
- Collection Cost
- Development Cost
- Erection Cost
- Management Cost

Index	Project ID with serial	Number of turbines	Turbine rating MW	Rotor diameter m	Module	Type of cost	Cost per turbine	Cost per project	Cost per kW
0	ass4_1ea34	3	3.37	130	FoundationCost	Equipment rental	5806	17418	1.72285
1	ass4_1ea34	3	3.37	130	FoundationCost	Labor	305000	915000	90.5045
2	ass4_1ea34	3	3.37	130	FoundationCost	Materials	106379	319136	31.5663
3	ass4_1ea34	3	3.37	130	FoundationCost	Mobilization	9513.78	28541.3	2.82308
4	ass4_1ea34	3	3.37	130	SitePreparationCost	Materials	1820.52	5461.57	0.540214
5	ass4_1ea34	3	3.37	130	SitePreparationCost	Equipment rental	628.586	1885.76	0.186524
6	ass4_1ea34	3	3.37	130	SitePreparationCost	Labor	4565.05	13695.1	1.35461
7	ass4_1ea34	3	3.37	130	SitePreparationCost	Other	130305	390916	38.6663
8	ass4_1ea34	3	3.37	130	SitePreparationCost	Mobilization	4988.54	14965.6	1.48028
9	ass4_1ea34	3	3.37	130	SubstationCost	Other	0	0	0
10	ass4_1ea34	3	3.37	130	TransportCost	Other	200000	624000	61.7211
11	ass4_1ea34	3	3.37	130	GridConnectionCost	Other	476499	1.4295e+06	141.394
12	ass4_1ea34	3	3.37	130	CollectionCost	Equipment rental	4167.74	12503.2	1.23672
13	ass4_1ea34	3	3.37	130	CollectionCost	Labor	19477.7	58433	5.77972
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17	ass4_1ea34	3	3.37	130	DevelopmentCost	Labor	50000	150000	14.8368

- Equipment rental
- Labor
- Materials
- Mobilization
- Fuel
- Construction Permitting
- Project Management
- Bonding
- Engineering Foundation and Collections System (includes met mast)
- Site Facility
- Insurance

Note: The **cost of turbines** themselves and the **cost of land** should be added to your analyses.

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Project Financial Analysis



Basic Terms

Cash Flow (R_t).

- **Net balance of cash** moving into and out of a business at specific point in time.
- Three types: operating, investing, and financing.
- Problem: timing of costs and benefits in a cash flow series → discounting!

Inflation Rates (i).

- Rise in price levels through increase in available currency and credit without proportionate increase in available goods and services of equal quality
→ value of money changes over time
- Cost and revenue can be expressed in
 1. **“current” dollar**: actual CF in year it incurs, changes over time due to inflation
 2. **“constant” dollar**: would have been required if cost was paid in base year→ Difference: inflation (base year: cash flows are the same)

Time Point / Periods.

- Base year (year to which all cash flows are converted)
- Dollar Year (year to which base year results are converted and reported)
- Analysis period vs. investment's life span (often equal)



Basic Terms

Discount Rates.

“Time value is the price put on the time that an investor waits for a return on an investment. A dollar received today is worth more than a dollar received tomorrow because the dollar today can be invested to earn interest immediately. Conversely, a dollar received tomorrow is worth less than a dollar received today because the opportunity to earn interest on the dollar is lost.”

- Discount rate = **measure of time value**, essential to calculate present value
- Either includes effects of inflation (= **nominal discount rate** d_n , current dollar analysis) or excludes it (= **real discount rate** d_r , constant dollar analysis)
Consistency throughout analysis!

$$(1 + d_n) = (1 + d_r) \cdot (1 + i)$$

- Not the same for all investors! Depend on rate of return, risk premium, planning horizon, interest rates, ...
→ varies from state to state, industry to industry, and company to company.
- Cost of capital must be recovered by investor to warrant his investment.



Basic Terms

Present Value.

- measure of **today's value of revenues or costs in the future**

$$PV = \frac{F_n}{(1 + d)^n}$$

with F_n ...cash flow n years in the future; d ...annual discount rate

Annuity.

- When future cash flows are fixed in size and regularly occur over a specific number of periods

$$PV = F \cdot \frac{(1 + d)^N - 1}{d(1 + d)^N} = F \cdot A$$

with F ...cash flow in each of N future years and Annuity A :

$$A = \frac{(1 + d)^N - 1}{d(1 + d)^N}$$



Economic Measures

Net Present Value.

- One way to examine costs (cash outflows) and revenues (cash inflows) together
- Evaluate investment features / decision
- Form of streams (current or constant dollars) must be known to apply correct discount rate

$$NPV = \sum_{n=0}^N \frac{F_n}{(1+d)^n} = F_0 + \frac{F_1}{(1+d)^1} + \frac{F_2}{(1+d)^2} + \dots + \frac{F_N}{(1+d)^N}$$

with F_n ... net cash flow in year n ; N ... analysis period

Total Life-Cycle Cost.

- Costs incurred through ownership of an asset over its life span / period of interest
- Considers all significant costs over life of the project discounted to base year

$$TLCC = \sum_{n=0}^N \frac{C_n}{(1+d)^n}$$

with C_n ...cost in period n : investment costs, expected salvage value, O&M and repair, replacement, energy



Economic Measures

Total Life-Cycle Cost (cont.).

Simple example with nominal discount rate = 12%

Project investment = \$10,000

O&M cost in year zero = \$1,300, inflating with $i = 3\%$ per year thereafter

No salvage value, investment is not replaced, no taxes

Current Dollar TLCC Evaluation

Year	Investment	Discounted Investment	Current Dollar O&M Costs	Discounted O&M Costs
0	\$ 10,000	\$ 10,000	\$ 0	\$ 0
1	0	0	1,339	1,196
2	0	0	1,379	1,100
3	0	0	1,421	1,011
4	0	0	1,463	930
5	0	0	1,507	855
NPV		\$ 10,000		\$ 5,091
TLCC = \$ 15,091				

nominal
discount rate

Economic Measures

Total Life-Cycle Cost (cont.).

Simple example with nominal discount rate = 12%

Project investment = \$10,000

O&M cost in year zero = \$1,300, inflating with $i = 3\%$ per year thereafter

No salvage value, investment is not replaced, no taxes

Constant Dollar TLCC Evaluation

Year	Investment	Discounted Investment	Constant Dollar O&M Costs	Discounted O&M Costs
0	\$ 10,000	\$ 10,000	\$ 0	\$ 0
1	0	0	1,300	1,196
2	0	0	1,300	1,100
3	0	0	1,300	1,011
4	0	0	1,300	930
5	0	0	1,300	855
NPV		\$ 10,000		\$ 5,091
TLCC = \$ 15,091				

recall:

$$(1 + d_n) = (1 + d_r) \cdot (1 + i)$$

→

$$d_r = \frac{1 + d_n}{1 + i} - 1$$

→

$$d_r = 8.74\%$$

real discount rate

Economic Measures

Levelized Cost of Energy.

“The LCOE is that cost hat, if assigned to every unit of energy produced by the system over the analysis period, will equal the TLCC when discounted back to the base year.”

$$\sum_{n=0}^N \frac{Q_n \cdot LCOE}{(1+d)^n} = TLCC$$

$$LCOE = \frac{TLCC}{\sum_{n=0}^N \frac{Q_n}{(1+d)^n}}$$

with Q_n ...energy output in year n , N ...analysis period

If the system output Q remains constant over time and considering one year as time period hence $Q = AEP$ (Annual Energy Production), the equation reduces to

$$LCOE = \frac{TLCC}{AEP \cdot A}$$

recall: A ...Annuity



Economic Measures

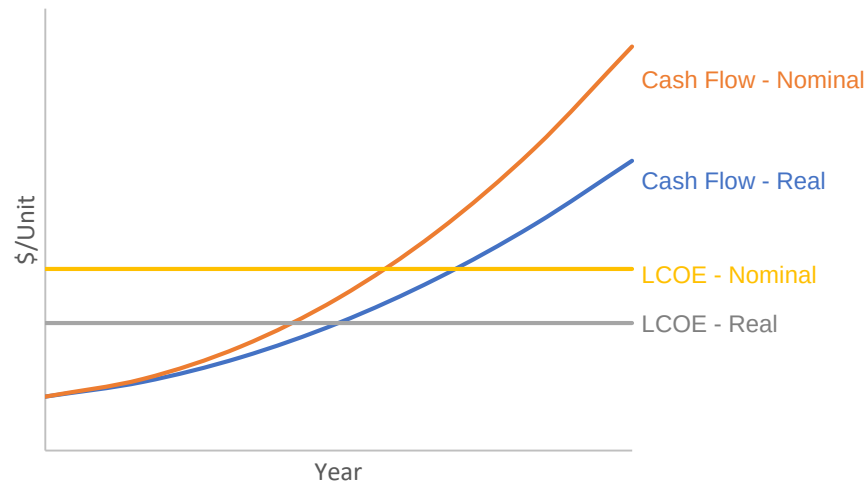
Levelized Cost of Energy (Cont.).

Neglecting taxes, assuming constant O&M costs per year, no financing, constant inflation rate, constant AEP, and no salvage value, we can write:

$$LCOE = \frac{I}{AEP \cdot A} + \frac{O\&M}{AEP}$$

with I...Initial Investment

$A = f(d) \rightarrow$ discount rate in A can be nominal or real, depending on the analysis purpose. Typically: Short-term studies in current dollars, long-term studies in real dollars



Economic Measures

Levelized Cost of Energy (Cont.).

Coming back to our **example** with TLCC = \$ 15,091

Assuming AEP = 1000 units/a

$d = d_n = 12\%$

$$A = \frac{(1 + d)^N - 1}{d(1 + d)^N} = 3.60 \text{ a}$$

$$LCOE = \frac{TLCC}{AEP \cdot A} = \frac{\$15,091}{1000 \frac{\text{units}}{\text{a}} \cdot 3.60 \text{ a}} = 4.19 \frac{\$}{\text{unit}}$$

Selling energy for \$4.19 per unit in current dollars over the next 5 years

→ **investor recoups initial investment and annual O&M costs, and earns 12%**

Repeating the exercise for constant dollars → LCOE = \$3.85 per unit
(→ ... and earns 8.74%)



Economic Measures

So far: cost-based

Now: **value-based** (considering earnings through selling energy)

For a project to be profitable, the total earnings must be higher than the total cost.

Internal Return Rate (IRR).

= discount rate that makes the net present value (NPV) of all negative and positive cash flows equal to zero in a discounted cash flow analysis.

$$0 = NPV = \sum_{n=0}^N \frac{F_n}{(1 + IRR)^n}$$

Solutions: roots of polynomial of n^{th} order

→ For higher order: graphically or numerically



Economic Measures

Internal Return Rate (cont.).

Year	Investment	Current Dollar O&M Costs	Revenues	Cash flow	Discounted Cash flow
0	\$ 10,000	\$ 0	\$ 0	\$-10,000	\$-10,000
1	0	1,339	4,120	2,781	...
2	0	1,379	4,244	2,865	...
3	0	1,421	4,371	2,950	...
4	0	1,463	4,502	3,039	...
5	0	1,507	4,637	3,130	...

function
of d_n

Coming back to our example.

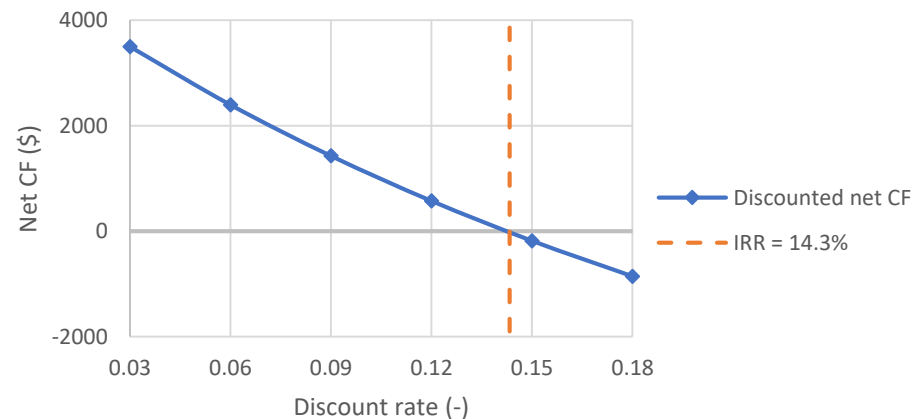
Energy price:

$P = \$4$ per unit in year 0

Inflating at 3% per year

→ Here: graphical solution

→ up to $d_n = 14.3\%$, the project is profitable



Economic Measures

Profitability Index (PI).

$$PI = \frac{\sum_{n=1}^N \frac{F_n}{(1+d)^n}}{I} = 1 + \frac{NPV}{I}$$

- Compares payoff (PV of future cashflows) with initial investment
- Measure of a project's or investment's attractiveness
- To be profitable: $PI \geq 1.0$

In our example with $d_n = 12\%$:

$$PI = \frac{\$10,574}{\$10,000} = 1.06 \geq 1$$

→ profitable

Year	Discounted Cash flow
0	\$-10,000
1	2483
2	2284
3	2100
4	1931
5	1776

